

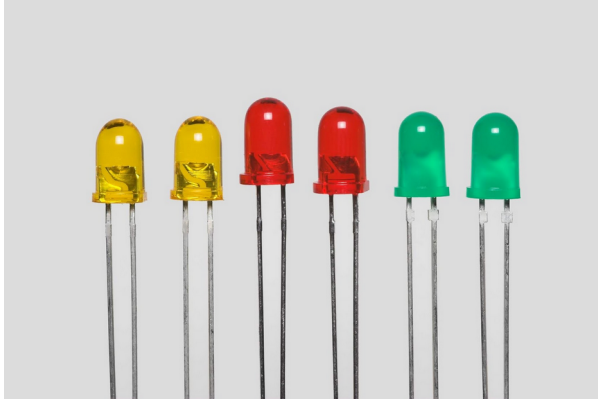
Basic Electronics and Microcontrollers

Electronics

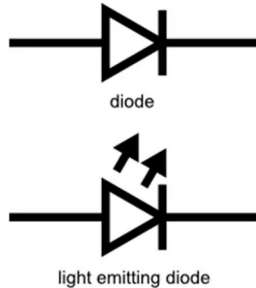
Electronic Components

- LED's / Diodes
- Buttons / Switches
- Sensors
- Resistors
- Transistors
- Capacitors
- Motors
- Relay
- Displays
- Breadboards
- IC
- Battery
- Transformer
- Inductor
- Thermistor
- Light Dependent Resistor
- Potentiometer

LEDs / Diodes

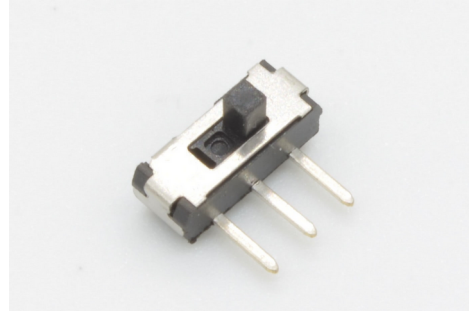
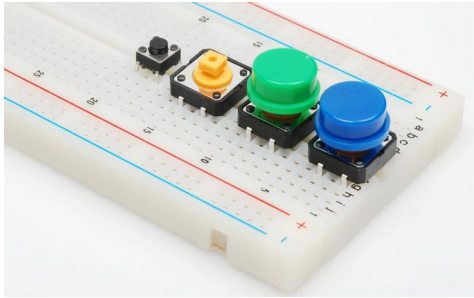


LEDs – Light Emitting Diode is a component that gives off light when electricity is passed through it.

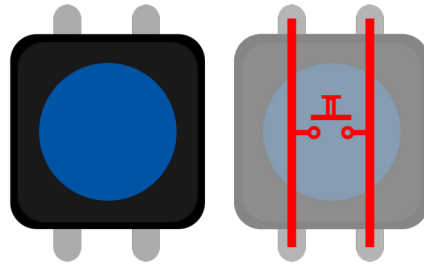


Diode – The diode and including the LED only allows electricity to pass through in 1 direction

Buttons and Switches

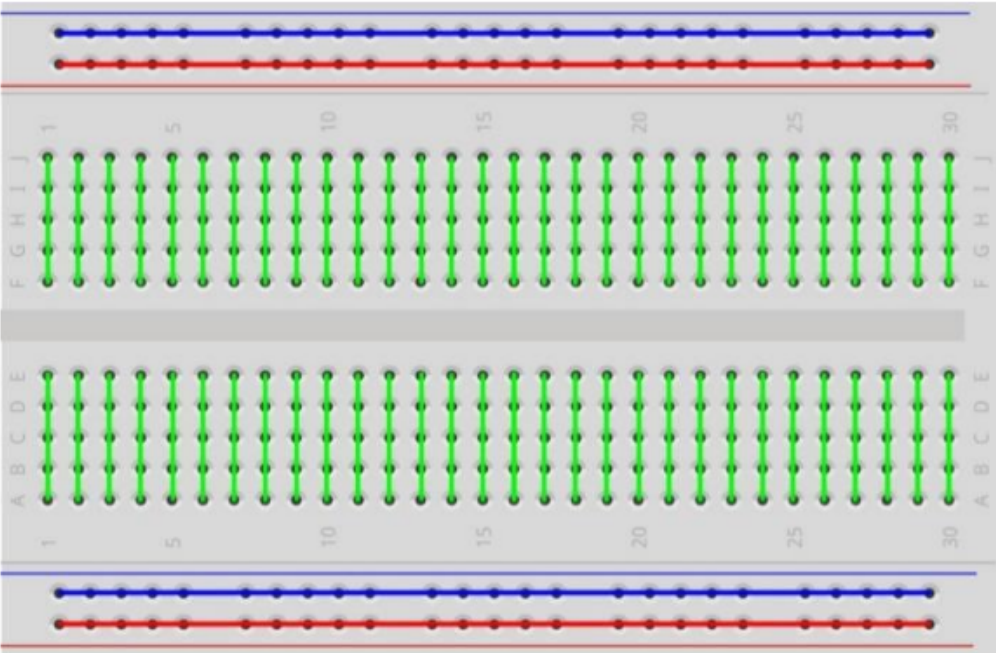


Switch – A component that turns on/off the flow of electricity by opening and closing the circuit



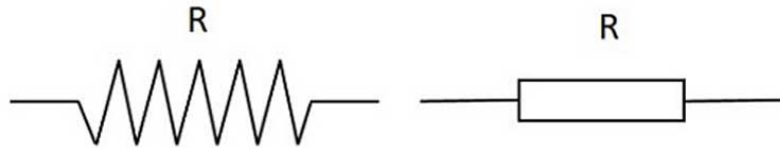
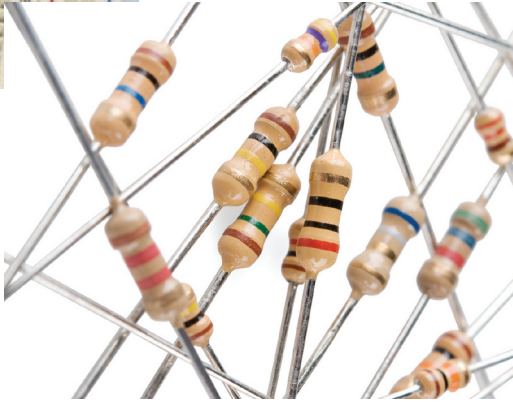
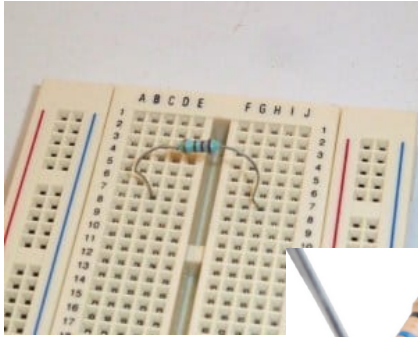
Buttons – Act like a switch but normally have a spring inside to pop the circuit open after you have let go.

Breadboards



Breadboards are reusable platform to build electronic circuits on without soldering, these are great for prototyping projects.

Resistors

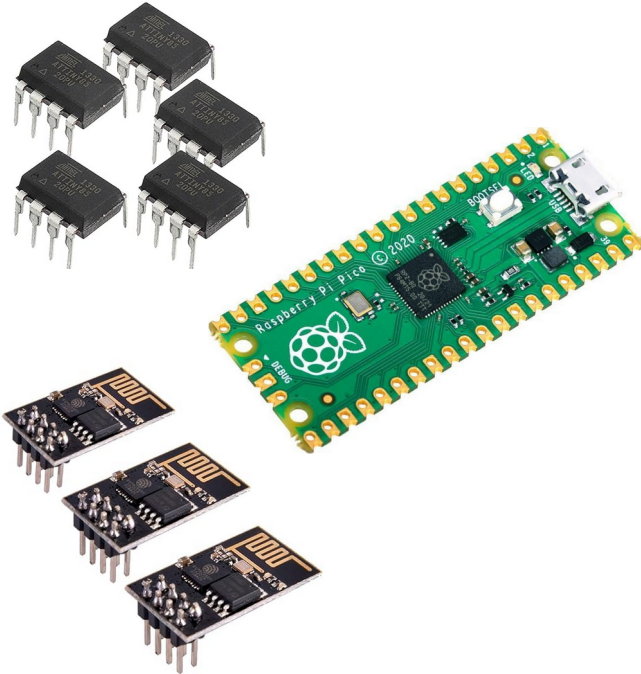


Symbol for a fixed resistor

Resistors – Restrict the current of electricity flow, these are commonly used with LEDs so they don't burn out.

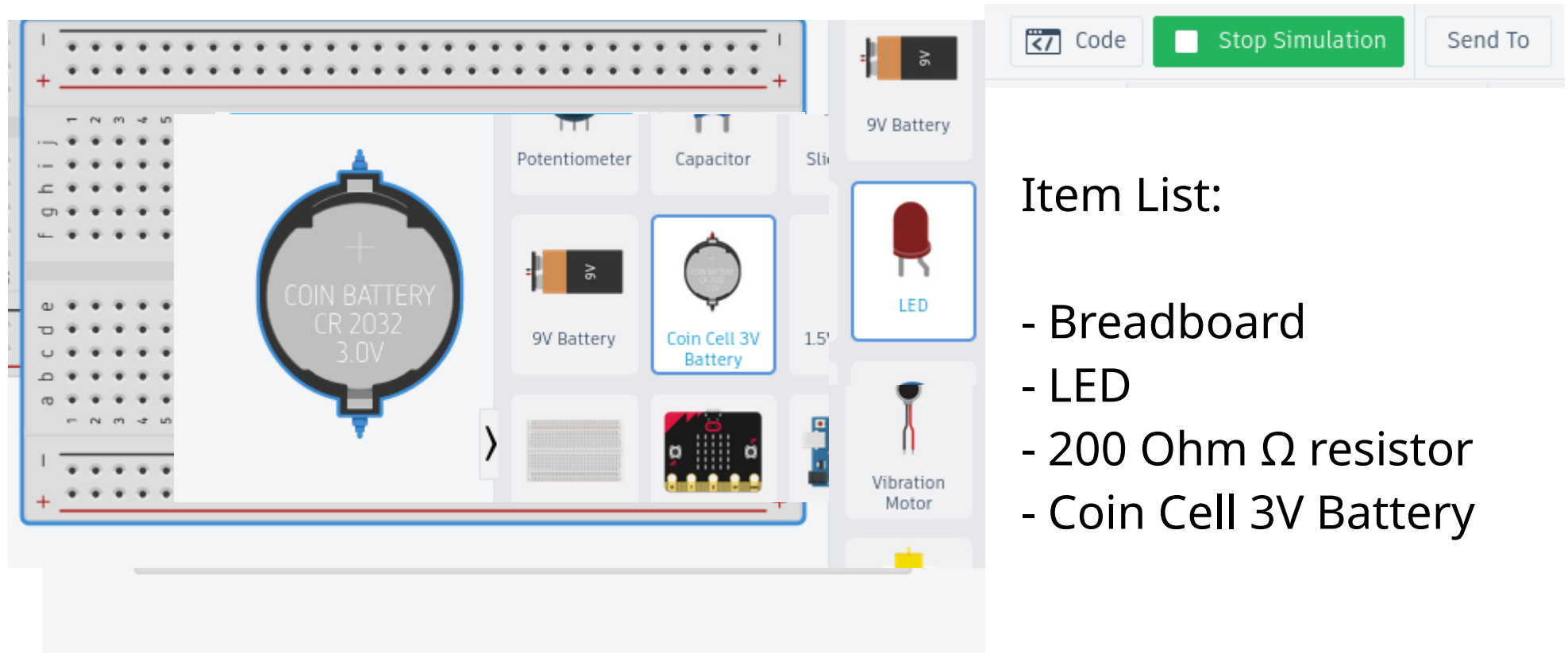
Think of this like a tap turning the flow of water down, more resistance less water.

Microcontrollers



A Microcontroller is a small computer on a single integrated circuit, It contains a small processor and programmable inputs/outputs and a small flash memory to store programs that run when powered up.

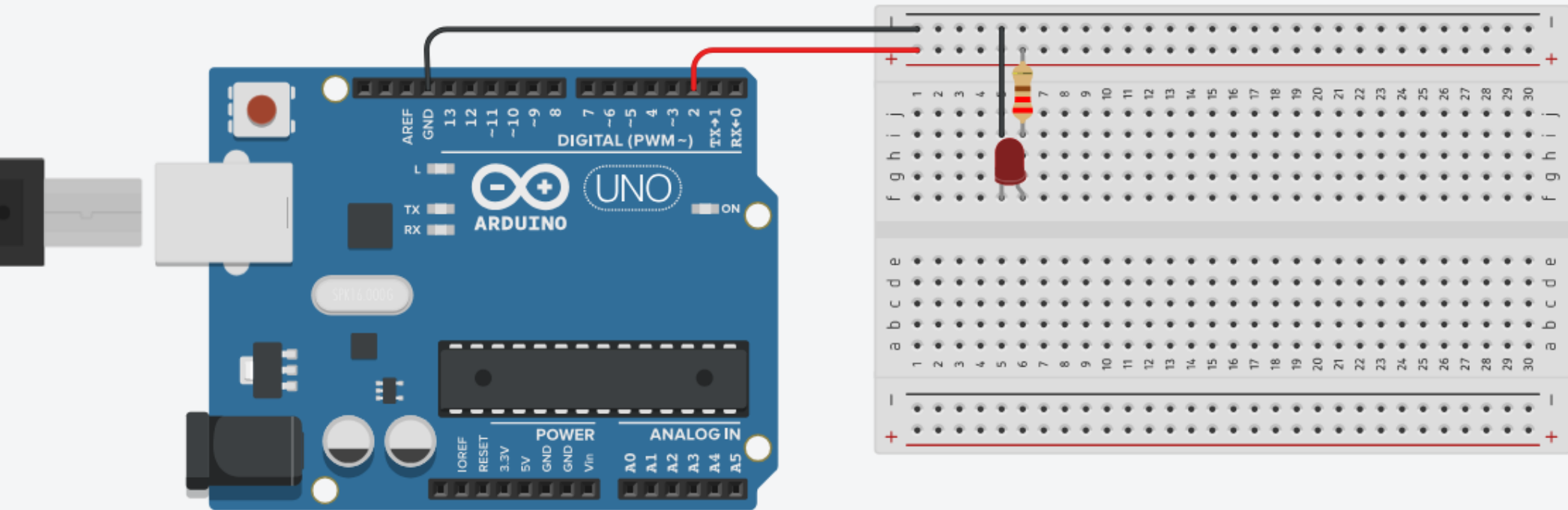
First Circuit



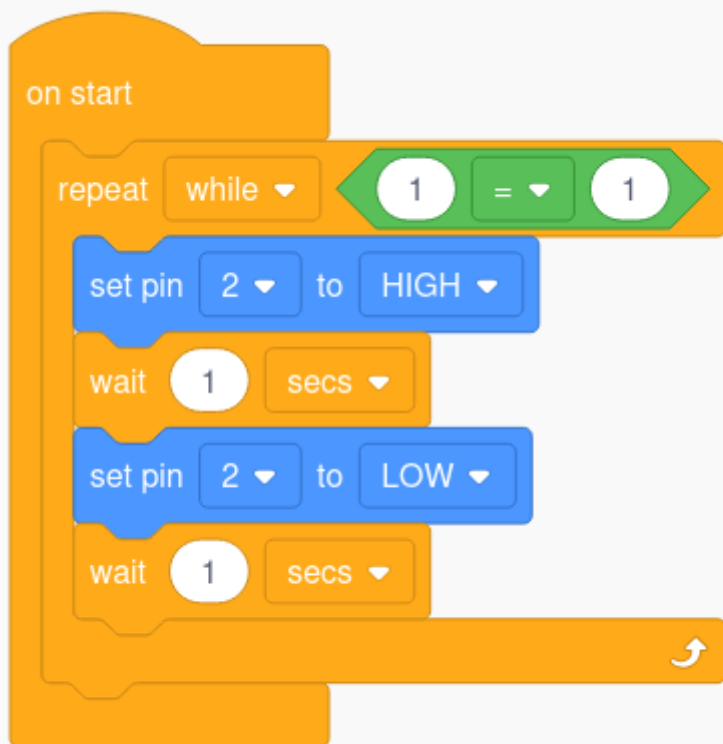
Item List:

- Breadboard
- LED
- 200 Ohm Ω resistor
- Coin Cell 3V Battery

Blinking LED



Code



```
1 // C++ code
2 //
3 void setup()
4 {
5     pinMode(2, OUTPUT);
6     while (1 == 1) {
7         digitalWrite(2, HIGH);
8         delay(1000); // Wait for 1000 millisecond(s)
9         digitalWrite(2, LOW);
10        delay(1000); // Wait for 1000 millisecond(s)
11    }
12 }
13
14 void loop()
15 {
16     delay(10); // Delay a little bit to improve simulation performance
17 }
```

Button Buzzer

